20. SSH Basics: Keys, Config, Jump Hosts

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Phase: Phase 9 — Security and SSH
Estimated time: 1–2 days
Slug: `20-ssh-basics`

Learning objectives

- Generate and use SSH keys
- Write ~/.ssh/config Host entries
- Disable password auth conceptually
- Use jump hosts / ProxyJump

Prerequisites

Complete prior numbered lessons. Prefer a real Linux VM/container for Linux-specific topics.

```
ssh-keygen -t ed25519 -C "you@example"
ssh-copy-id user@server
ssh user@server
```

Config

```
Host lab
  HostName 203.0.113.10
  User ada
  IdentityFile ~/.ssh/id_ed25519
  ProxyJump bastion
```

Prefer keys over passwords. Protect private keys (permissions 600). Never commit them.

Common mistakes

- Sharing private keys
- chmod 777 on .ssh
- Password auth on internet-facing SSH

Exercises

Work in `exercises/20-ssh-basics/`. Run:

```
./test.sh
```

- **A (Drill):** make `./test.sh` pass
- **B (Apply):** note one real command you ran on a Linux host related to this lesson
- **C (Challenge):** rewrite your solution from memory the next day

Do not open `solutions/20-ssh-basics/` until you have an honest failing attempt.

Self-check

1. ed25519 purpose?
2. What is ProxyJump?
3. Recommended .ssh private key perms?

Further reading

- https://www.openssh.com/manual.html